

AI Multi-Paper Knowledge Synthesizer

End-to-End System Data Flow, RAG Retrieval Engine, Model Stack & Setup Guide

1. Executive Overview

The AI Multi-Paper Knowledge Synthesizer is a full-stack research paper synthesis platform. It enables researchers to query across multiple AI manuscripts, execute dense vector retrieval using BAAI/bge-large-en-v1.5 1024-dimensional embeddings, generate synthesis reports with verifiable inline citations [C1], [C2], build publication-ready literature reviews via Nvidia Nemotron / Qwen LLM, interact with a research paper chatbot, and explore comparative methodological matrices & dynamic knowledge graph networks.

2. Core System Architecture Components

Frontend UI	React 19, TypeScript, Vite 6, Tailwind CSS v4, Lucide React, Framer Motion
Backend Servers	Express Node.js API (Port 3000) + Optional Python FastAPI Engine (Port 8000)
Embedding Model	BAAI/bge-large-en-v1.5 (1024-dimensional dense float vector representations)
Vector Database	Qdrant Vector DB with Cosine Similarity & BM25 Reciprocal Rank Fusion
LLM Reasoning	Nvidia Nemotron 70B (Cloud API) / Qwen 2.5 3B (Local Ollama Engine)

3. End-to-End Data Flow Pipeline (Step-by-Step)

Step 1: Input & Ingestion Layer

Accepts live ArXiv API search queries (<https://export.arxiv.org/api/query>) or user PDF/Text uploads. XML and PDF-parse extract paper titles, authors, published dates, abstracts, and raw text text buffers.

Step 2: Text Parsing & Recursive Chunking

Manuscripts are split recursively using hierarchical separators ["\n\n", "\n", ". ", " "] into 700-character chunks with 100-character overlap. Each chunk is tagged with paper metadata payloads.

Step 3: Dense Vector Embedding & Qdrant Storage

1024-dimensional dense float vector embeddings are computed for each chunk using BAAI/bge-large-en-v1.5 model. Vector points and metadata are indexed in Qdrant Vector DB.

Step 4: Hybrid RAG Search & Citation Mapping

Retrieves top-K chunks using Hybrid Score = 0.65 * CosineSimilarity + 0.35 * BM25. Chunks are mapped into verifiable evidence blocks tagged with inline citations [C1], [C2], [C3].

Step 5: LLM Inference & React UI Output

Nvidia Nemotron / Qwen 2.5 LLM processes the grounded prompt and produces markdown responses, structured JSON matrices, dynamic SVG Knowledge Graph networks, and PDF report downloads.

4. Local Hardware & Execution Specification

Optimized Configuration for AMD Ryzen 5 3500U / 12 GB RAM Specs:

- Recommended Local LLM: qwen2.5:3b (Ollama Engine - 1.9 GB RAM footprint, ~20 tok/sec)
- Local Vector Embedding: BAAI/bge-large-en-v1.5 via Python sentence-transformers / Transformers.js
- Offline Fallback: Zero-latency built-in RAG reasoning engine with offline paper paragraph mapping